

Figure 1: Python download page from the official portal

Figure 2 depicts the execution of Python code on a system in which Python 2.7 is installed in Drive E: of the Windows OS.

IDE based programming with Python can include any IDE to write, debug and execute the code.

Given below is a list of Python IDEs where a graphical user interface is provided for easy programming:

IDE	IntelliJ IDEA
Koding	Anjuta
Eric	Geany
Komodo IDE	Ninja-IDE
PIDA	KDevelop
MonoDevelop	PyCharm
Spyder	PyDev
PyScripter	SourceLair
Stani's Python Editor	Python Tools for Visual Studio
PythonAnywhere	Pyzo
Understand	Thonny

Digital forensics using Python programming

Whenever the topics of digital forensics, cyber security and penetration testing are discussed, professionals generally depend on a number of third party tools and operating systems. Kali Linux, Metasploit, Parrot Security OS and many other tools are used for digital forensics. These tools come with in-built applications which the users deploy without real knowledge of the internal architecture and algorithmic approach of implementation.

Python is a widely used programming language for cyber security, penetration testing and digital forensic applications. Using the base programming of Python, any of the following can be performed without using any other third party tool:

- Web server fingerprinting
- Simulation of attacks
- Port scanning
- Website cloning
- Load generation and testing of a website
- Creating intrusion detection and prevention systems
- Wireless network scanning



Figure 2: Executing Python code at the Windows command shell interface

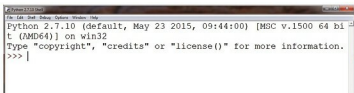


Figure 3: Python IDLE environment

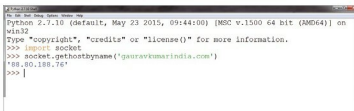


Figure 4: Fetching an IP address of a website from a URL



Figure 5: Fetching IP addresses associated with the local system

- Transmission of traffic in the network
- Accessing mail servers...
- ...and many other implementations related to digital fingerprinting and security applications

Socket programming

Socket programming is in-built with Python, similar to Java. To work with socket programming, the package socket is first imported and then the related methods can be called. Python installation comes with the in-built IDLE GUI.

Network port scanning

Generally, the *nmap* tool is used for the implementation of network port scanning, but using Python socket programming, it can be implemented without any third party tool. In Kali Linux, there are many tools available for digital forensics related to networks, but many of these implementations can be done using Python programming with just a few lines of instruction.

The code for port scanning of any IP address can be downloaded from http://opensourceforu.com/article_source_code/sept16/digital_forensic.zip. The code checks which particular ports are open from the PortList [20, 22, 23, 80, 135, 445, 912]. Each value in the PortList specifies a